

Packet Tracer – OSPF Basic Configuration and Multi-Area Routing

Objectives

Part 1: Build a Multi-Router OSPF Topology

Part 2: Configure IP Addressing

Part 3: Configure Single-Area OSPF

Part 4: Configure Multi-Area OSPF

Part 5: Verify OSPF Neighbor Relationships

Part 6: Verify Routing Tables and Test Connectivity

Background / Scenario

As networks continue to grow, routing protocols must scale efficiently while quickly adapting to network changes. Unlike RIP, which uses hop count as its metric, **Open Shortest Path First (OSPF)** uses cost-based routing and converges much faster.

In this lab, you will configure OSPF on three routers, establish OSPF neighbor relationships, and verify route exchange. You will also introduce the concept of **multiple OSPF areas**, a key feature used in enterprise networks.

This lab focuses on:

- Configuring OSPF
- Understanding OSPF Areas
- Verifying OSPF Neighbor Adjacencies
- Examining OSPF Routing Tables
- Testing end-to-end connectivity

You are taking your first step into enterprise-grade routing protocols commonly used in modern networks.

Instructions

Part 1: Build the Network Topology

Step 1: Add the Devices

a. Open **Cisco Packet Tracer**.

b. Add the following devices:

- 3 Routers (1941 or 2911)
- 2 Switches (2960)
- 2 PCs

c. Rename the devices:

- Router0 → **R1**
 - Router1 → **R2**
 - Router2 → **R3**
 - Switch0 → **S1**
 - Switch1 → **S2**
 - PC0 → **PC-A**
 - PC1 → **PC-B**
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Step 2: Connect the Devices

Use **Copper Straight-Through** cables.

Device Interface		Device Interface	
PC-A	FastEthernet0	S1	Fa0/1
S1	Gi0/1	R1	Gi0/0
R1	Gi0/1	R2	Gi0/0
R2	Gi0/1	R3	Gi0/0
R3	Gi0/1	S2	Gi0/1
S2	Fa0/1	PC-B	FastEthernet0

Wait until all interfaces become green.

Network Addressing Plan

Device Interface IP Address

PC-A	Fa0	192.168.1.10 /24
R1	Gi0/0	192.168.1.1 /24
R1	Gi0/1	10.0.12.1 /30
R2	Gi0/0	10.0.12.2 /30
R2	Gi0/1	10.0.23.1 /30
R3	Gi0/0	10.0.23.2 /30
R3	Gi0/1	192.168.3.1 /24
PC-B	Fa0	192.168.3.10 /24

Part 2: Configure IP Addressing

Step 1: Configure PC-A

Navigate to:

Desktop → IP Configuration

Enter:

- IP Address: 192.168.1.10
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.1.1
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Step 2: Configure PC-B

Enter:

- IP Address: 192.168.3.10

- Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.3.1
-

Step 3: Configure Router Interfaces

R1

enable

configure terminal

interface g0/0

ip address 192.168.1.1 255.255.255.0

no shutdown

exit

interface g0/1

ip address 10.0.12.1 255.255.255.252

no shutdown

exit

R2

enable

configure terminal

interface g0/0

ip address 10.0.12.2 255.255.255.252

no shutdown

exit

interface g0/1

ip address 10.0.23.1 255.255.255.252

no shutdown

exit

R3

```
enable
configure terminal
```

```
interface g0/0
ip address 10.0.23.2 255.255.255.252
no shutdown
exit
```

```
interface g0/1
ip address 192.168.3.1 255.255.255.0
no shutdown
exit
```

Part 3: Configure Single-Area OSPF

What is OSPF?

OSPF (Open Shortest Path First) is a dynamic routing protocol that:

- Uses Cost instead of Hop Count
- Converges faster than RIP
- Supports large networks
- Uses Areas for scalability
- Exchanges Link-State Advertisements (LSAs)

Unlike RIP, OSPF creates neighbor relationships before exchanging routing information.

Step 1: Configure OSPF on R1

```
router ospf 1

network 192.168.1.0 0.0.0.255 area 0
network 10.0.12.0 0.0.0.3 area 0
```

Step 2: Configure OSPF on R2

```
router ospf 1
```

```
network 10.0.12.0 0.0.0.3 area 0
```

```
network 10.0.23.0 0.0.0.3 area 0
```

Step 3: Configure OSPF on R3

```
router ospf 1
```

```
network 10.0.23.0 0.0.0.3 area 0
```

```
network 192.168.3.0 0.0.0.255 area 0
```

Understanding the Wildcard Mask

OSPF uses wildcard masks instead of subnet masks.

Examples:

Subnet Mask	Wildcard Mask
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255.255.255.0	0.0.0.255
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255.255.255.252	0.0.0.3
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A wildcard mask is simply the inverse of the subnet mask.

Part 4: Verify OSPF Neighbor Relationships

Step 1: View OSPF Neighbors

On R2:

```
show ip ospf neighbor
```

Expected output:

Neighbor ID	State
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1.1.1.1	FULL
---------	------

3.3.3.3	FULL
---------	------

The **FULL** state indicates a successful OSPF adjacency.

Step 2: Verify OSPF Interfaces

show ip ospf interface brief

Expected output:

Interface	Area
Gi0/0	0
Gi0/1	0

Part 5: Verify OSPF Routes

Step 1: View Routing Table

On R1:

show ip route

Expected output:

O 10.0.23.0/30
O 192.168.3.0/24

Routes beginning with:

O

indicate routes learned through OSPF.

Step 2: Verify Routes on R3

show ip route

Expected output:

O 192.168.1.0/24
O 10.0.12.0/30

Part 6: Configure a Multi-Area OSPF Design

Enterprise networks commonly use multiple OSPF areas.

Modify the design as follows:

Area Assignments

Router	Network	Area
R1 LAN	192.168.1.0	Area 1
R1-R2 Link	10.0.12.0	Area 0
R2-R3 Link	10.0.23.0	Area 0
R3 LAN	192.168.3.0	Area 2

Reconfigure R1

```
router ospf 1
```

```
network 192.168.1.0 0.0.0.255 area 1
```

```
network 10.0.12.0 0.0.0.3 area 0
```

Reconfigure R3

```
router ospf 1
```

```
network 10.0.23.0 0.0.0.3 area 0
```

```
network 192.168.3.0 0.0.0.255 area 2
```

Why Area 0?

Area 0 is called the Backbone Area.

All other OSPF areas must connect to Area 0 either directly or through virtual links.

Part 7: Test End-to-End Connectivity

Step 1: Ping Between PCs

On PC-A:

ping 192.168.3.10

Expected result:

Reply from 192.168.3.10

You should receive four successful replies.

Step 2: Ping Back

On PC-B:

ping 192.168.1.10

Connectivity should succeed.

Step 3: Trace the Route

On PC-A:

tracert 192.168.3.10

Expected path:

192.168.1.1

10.0.23.2

192.168.3.10

Troubleshooting Tips

If OSPF is not working:

✓ Verify interfaces are up:

show ip interface brief

✓ Verify OSPF neighbors:

show ip ospf neighbor

✓ Verify OSPF routes:

show ip route

✓ Verify OSPF networks:

show running-config

✓ Ensure area numbers match on both sides of a link.

✓ Confirm IP addressing is correct.

✓ Check for FULL neighbor state.

What You Learned

- How OSPF differs from RIP.
 - How to configure OSPF using network statements.
 - How OSPF neighbor relationships are formed.
 - How to verify OSPF routes.
 - How to interpret OSPF routing tables.
 - How OSPF areas improve scalability.
 - The purpose of Area 0 (Backbone Area).
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Challenge Extensions (Optional)

Try modifying the lab:

1. Add a fourth router in Area 2.
2. Configure Router IDs manually:

```
router ospf 1
```

```
router-id 1.1.1.1
```

3. Use Simulation Mode to observe OSPF Hello packets.
4. Shut down a link and observe OSPF reconvergence.
5. Compare OSPF routes with RIP routes from the previous lab.
6. Configure passive interfaces for LAN networks.